Nonlinear model-based control instead cancels known rigid-body dynamics [Slotine 91] estimates uncertain parameters online [Åström 95], or bounds their effect. Yang et al. [Yang 26], for example, combine state constraints with explicit valve dead-band inversion. Such methods can track accurately, but depend on valve, friction and inertia models that are costly to maintain on a commercial implement [Mattila 17].

This thesis therefore retains reliable structure and learns the uncertain part. The DLS kinematic controller acts as teacher and data-collection controller; learning covers the mapping from cylinder command to realized motion and the policy that issues that command (Section 2.7).

2.4 Data-Driven System Identification and Learned Dynamics

2.4.1 Learned Forward and Inverse Dynamics

Robot model learning distinguishes forward models, which predict the next state from state and command, from inverse models, which predict the command for a desired transition [Nguyen-Tuong 11]. Section 5.1 fixes both interfaces. One-step fit is insufficient when a forward model is recursively rolled out, because small errors compound [Nagabandi 18]. The models here are therefore trained with multi-step objectives. Probabilistic ensembles represent growing rollout uncertainty [Chua 18]; recurrent architectures instead retain unobserved pressure and valve history [Hochreiter 97, Cho 14], as used by the reduced-order model.

2.4.2 Closed-Loop Identifiability

Informative identification requires inputs that excite the plant independently of feedback [Ljung 99, Isermann 11]. In closed-loop data, command and state history are correlated; a flexible predictor can score well by ignoring the command, and validation from the same distribution will not reveal the shortcut [Schoukens 19].

That failure occurred here: a closed-loop forward model reached one-step $R^2 \approx 0.999$ yet showed almost no command dependence under ablation. Chapter 4 therefore uses open-loop valve excitation, and every learned model is checked with zeroed and shuffled commands. Prediction score alone is not treated as evidence of control authority.

2.5 Reduced-Order Modeling

Reduced-order models retain dominant behavior at lower cost [Benner 15]. Here the full Simscape plant is reduced to per-cylinder extension, velocity and pressure difference. The construction is grey-box: known structure is embedded so that data fits what remains unknown [Karniadakis 21]. As in the structured dynamics network of Lutter et al. [Lutter 19], explicit physical layers improve interpretability. The hydraulic ROM retains the orifice, pressure and integration structure; learned terms represent valve realization, configuration-dependent load and bounded residuals.

Hwangbo et al. [Hwangbo 19] combined analytic rigid-body mechanics with a learned actuator response, a pattern later scaled in GPU-parallel simulation [Rudin 22]. Yamamori